

Right Triangles

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **apply** the Pythagorean theorem to determine a missing side.

RELATED SKILLS

I Apply the Pythagorean theorem or its converse.

A Extract perfect-power factors from a radical.

- ☐ 2. I can **use** the converse of the Pythagorean theorem to classify a triangle.

RELATED SKILLS

D Apply the Pythagorean theorem or its converse.

A Recognize and apply the converse of a conditional statement.

- ☐ 3. I can **apply** triangle-inequality relationships.

RELATED SKILLS

A Classify a triangle by side lengths and angle measures.

A Select a theorem that connects known facts to a target statement.

- ☐ 4. I can **determine** exact side lengths in 45-45-90 and 30-60-90 triangles.

RELATED SKILLS

I Recall special-right-triangle ratios.

A Distinguish exact values from measured or rounded approximations.

R Extract perfect-power factors from a radical.

- ☐ 5. I can **apply** right-triangle similarity relationships.

RELATED SKILLS

D Set up proportions using corresponding sides.

A Recognize AA triangle similarity.

- ☐ 6. I can **use** geometric-mean relationships created by an altitude to the hypotenuse.

RELATED SKILLS

I Apply geometric-mean relationships in a right triangle split by an altitude.

A Set up proportions using corresponding sides.

☐ **7. I can **select** sine, cosine, or tangent for a right-triangle problem.**

RELATED SKILLS

- ☐ I Label sides relative to a selected angle.
 - ☐ I Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
 - ☐ I Recall and correctly use the terms leg, hypotenuse, opposite side, adjacent side, and trigonometric ratio.
 - ☒ A Add congruence, parallel, perpendicular, and angle markings to a diagram.
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☐ **8. I can **determine** a missing side or angle of a right triangle.**

RELATED SKILLS

- ☐ I Use a trigonometric ratio to determine a missing side.
 - ☐ I Use an inverse trigonometric function to determine a missing angle.
 - ☒ A Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
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☐ **9. I can **solve** a contextual right-triangle problem.**

RELATED SKILLS

- ☐ D Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
 - ☒ A Restrict a model to meaningful contextual inputs.
 - ☒ A Use a trigonometric ratio to determine a missing side.
 - ☒ A Use an inverse trigonometric function to determine a missing angle.
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☐ **10. I can **justify** a right-triangle solution using an appropriate theorem or ratio.**

RELATED SKILLS

- ☐ D Select a theorem that connects known facts to a target statement.
 - ☒ A Apply the Pythagorean theorem or its converse.
 - ☒ A Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
 - ☒ A Recall special-right-triangle ratios.
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